

TASK 3 — DATA ANALYSIS & VISUALIZATION

3.1 Pass/Fail Distribution

```
plt.figure(figsize=(7,5))

sns.countplot(
    x="performance",
    data=df,
    palette="Set2"
)

plt.title("Student Performance Distribution")
plt.xlabel("Performance")
plt.ylabel("Number of Students")

plt.show()
```

Output

```
Pass
Fail
```

Analysis

The bar chart shows the number of students classified as Pass and Fail. This helps us understand the distribution of the target variable and determine whether the dataset is reasonably balanced.

3.2 Study Time vs Performance

```
plt.figure(figsize=(8,5))

sns.countplot(
    x="studytime",
    hue="performance",
```

```

        data=df,
        palette="Set1"
    )

    plt.title("Study Time vs Student Performance")
    plt.xlabel("Study Time")
    plt.ylabel("Number of Students")

    plt.show()

```

Output

A grouped bar chart showing Pass and Fail counts for different study-time categories.

Analysis

The visualization helps identify whether study time is associated with student performance. Students who spend more time studying may show a higher proportion of satisfactory performance.

3.3 Absences Histogram

```

plt.figure(figsize=(8,5))

sns.histplot(
    df["absences"],
    bins=20,
    kde=True,
    color="steelblue"
)

plt.title("Distribution of Student Absences")
plt.xlabel("Number of Absences")
plt.ylabel("Number of Students")

plt.show()

```

Output

A histogram showing the distribution of absences.

Analysis

The histogram shows how frequently different levels of absence occur among students. It allows us to identify whether most students have relatively low absences or whether high absence values are common.

3.4 Previous Failures vs Performance

```
plt.figure(figsize=(8,5))

sns.countplot(
    x="failures",
    hue="performance",
    data=df,
    palette="coolwarm"
)

plt.title("Previous Failures vs Student Performance")
plt.xlabel("Number of Previous Failures")
plt.ylabel("Number of Students")

plt.show()
```

Analysis

This visualization compares previous academic failures with the predicted performance category. Students with a history of previous failures may have a greater risk of poor academic outcomes.

3.5 Correlation Heatmap

```
numeric_columns = [
    "age",
    "studytime",
    "failures",
    "absences",
    "Medu",
    "Fedu",
    "goout",
    "health",
    "G1",
    "G2",
```

```
    "G3"  
]  
  
plt.figure(figsize=(12,8))  
  
correlation = df[numeric_columns].corr()
```

Analysis

The correlation heatmap shows the strength and direction of relationships between numerical variables. It helps identify variables that have stronger relationships with academic grades and can therefore be useful during feature analysis.

3.6 Grade Distribution

```
plt.figure(figsize=(8,5))  
  
sns.histplot(  
    df["G3"],  
    bins=20,  
    kde=True,  
    color="green"  
)  
  
plt.title("Distribution of Final Grades")  
plt.xlabel("Final Grade (G3)")  
plt.ylabel("Number of Students")  
  
plt.show()
```

Analysis

The histogram shows the distribution of final grades among students. It provides an overview of academic performance before converting the grades into Pass and Fail categories.



